

# Homework 7: Factor Analysis

S&DS 5360 | Multivariate Statistics

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## Part 1 | Setup & Data Overview

We continue our work with the Healthy Minds Survey (HMS) data. For Factor Analysis, we will select three distinct survey groups

- Depression (`phq9_1` to `phq9_9`): Nine questions on interest, mood, sleep, and energy levels over 2 weeks.
- Anxiety (`gad7_1` to `gad7_7`): Seven questions on nervousness, worrying, and restlessness over 2 weeks.
- Flourishing (`diener1` to `diener8`): Eight questions assessing positive psychological and social well-being.
- Resilience (`brs_1` to `brs_6`): Six questions measuring a student's ability to cope with stress and adversity. Note that we will need to reverse-score the even questions so that the scale remains consistent (higher score indicates higher resilience).

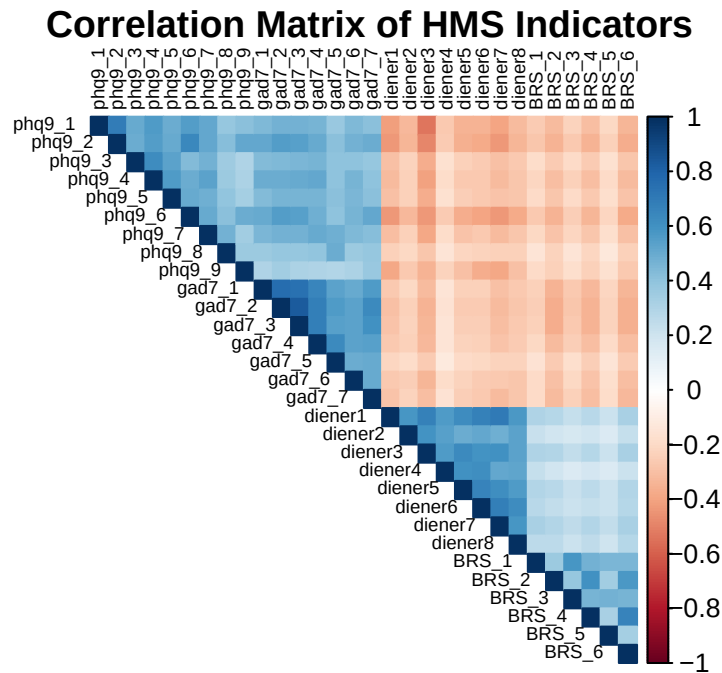
We omit the code to preprocess the data.

## Part 2 | Correlation Matrix

To better understand the relationship between our variables (the HMS survey questions), we begin by presenting a correlation matrix. We expect correlation, particularly among within-group questions (e.g., different questions in the `phq9` depression category should be correlated).

```
cormat <- cor(df)
corrplot(cormat, method = 'color', type = 'upper',
         tl.col = 'black', tl.cex = 0.6,
```

```
title = 'Correlation Matrix of HMS Indicators',
mar = c(0,0,1,0))
```



As expected, observe our within-scale groupings (the dark blue triangular blocks). The questions from **phq9**, **gad7**, and **diener** all exhibit intra-group correlations, indicating strong internal consistency within the scales. Turning our attention to the larger blue rectangles, these variables represent the comorbidity between depression and anxiety indicators (**phq9** and **gad7**, respectively). We have previously shown the interconnected relationship between these two question groupings.

Finally, we see a large red rectangle emerge between the flourishing (**diener**) and resilience (**BRS**) questions with those of **phq9** and **gad7**. We have previously observed this inverse relationship (or better said, negative correlation) in the past for the former: positive well-being and flourishing should be inversely related to psychological distress. It makes sense that resilience would follow this same relationship.

## Part 3 | KMO for Suitability of Data for Analysis

To determine the suitability of our data for factor analysis, we calculate the Kaiser-Meyer-Olkin (KMO) measure of sampling adequacy using the **EFATools** package introduced in class.

```
KMO(as.matrix(df))
```

Kaiser-Meyer-Olkin factor adequacy

Call: KMO(r = as.matrix(df))

Overall MSA = 0.96

MSA for each item =

|         |         |         |         |        |        |         |         |         |         |
|---------|---------|---------|---------|--------|--------|---------|---------|---------|---------|
| phq9_1  | phq9_2  | phq9_3  | phq9_4  | phq9_5 | phq9_6 | phq9_7  | phq9_8  | phq9_9  | gad7_1  |
| 0.96    | 0.96    | 0.96    | 0.96    | 0.98   | 0.98   | 0.98    | 0.96    | 0.97    | 0.97    |
| gad7_2  | gad7_3  | gad7_4  | gad7_5  | gad7_6 | gad7_7 | diener1 | diener2 | diener3 | diener4 |
| 0.94    | 0.95    | 0.97    | 0.95    | 0.98   | 0.98   | 0.96    | 0.95    | 0.96    | 0.95    |
| diener5 | diener6 | diener7 | diener8 | BRS_1  | BRS_2  | BRS_3   | BRS_4   | BRS_5   | BRS_6   |
| 0.96    | 0.95    | 0.96    | 0.97    | 0.92   | 0.95   | 0.92    | 0.92    | 0.94    | 0.94    |

As seen above, the overall KMO score for our 30 variables is “marvellously” high at 0.959. Further, note that the individual scores for each variable exceed 0.91; thus, there is substantial shared variance among the questions, making our data highly suitable for factor extraction.

## Step 4 | Determine Latent Factors

Next, to determine the appropriate number of latent factors to use for our analysis, we use both parallel analysis and our traditional scree plot and eigenvalue interpretation from principal components analysis.

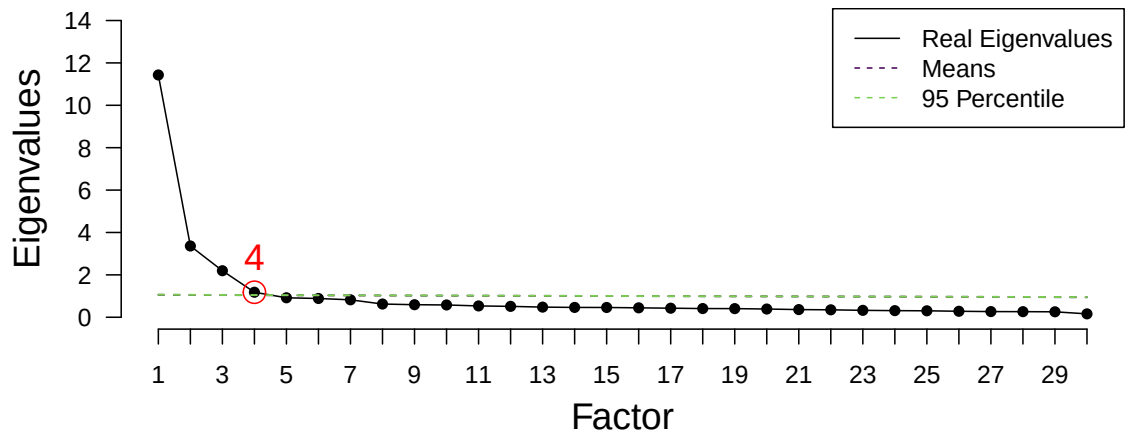
```
# Parallel Analysis  
PARALLEL(df)
```

Parallel Analysis performed using 1000 simulated random data sets  
Eigenvalues were found using PCA, SMC, and EFA

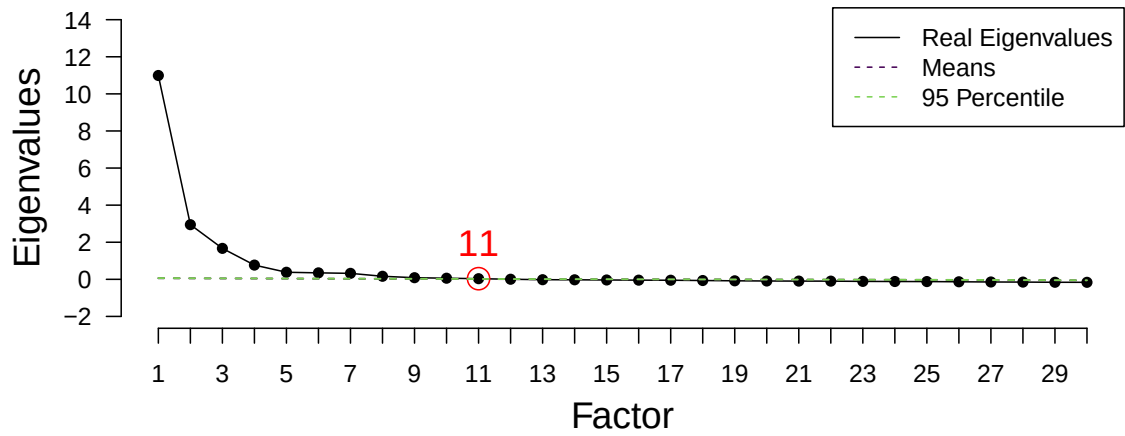
Decision rule used: means

```
-- Number of factors to retain according to -----  
  
* PCA-determined eigenvalues: 4  
* SMC-determined eigenvalues: 11  
* EFA-determined eigenvalues: 7
```

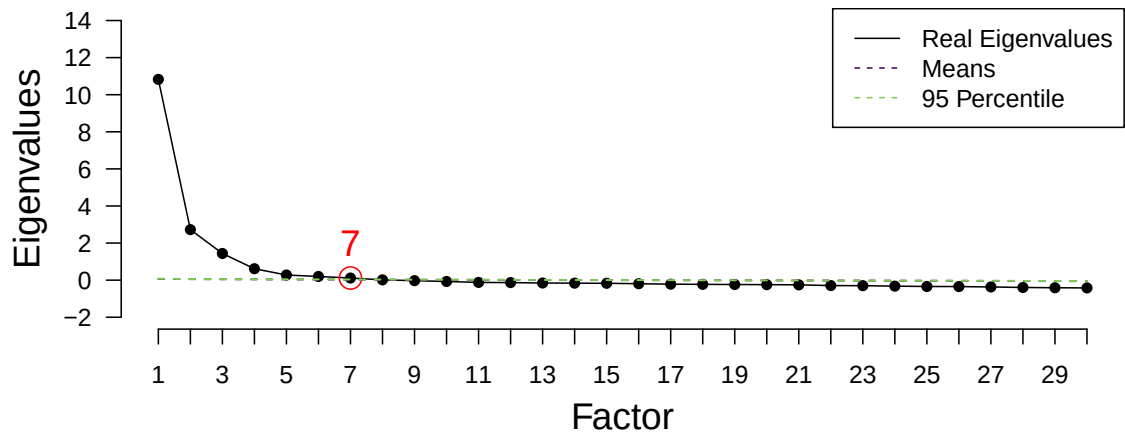
N Factors with Decision Rule 'means' and Eigen Type 'PCA': 4



N Factors with Decision Rule 'means' and Eigen Type 'SMC': 11



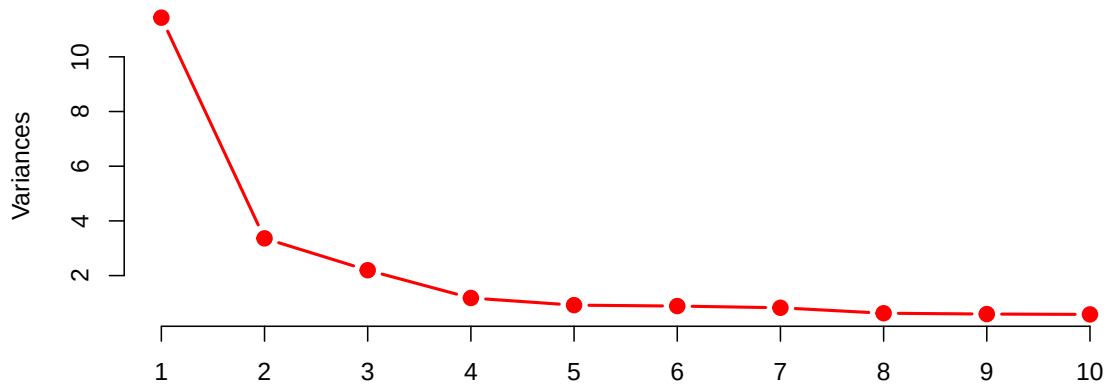
### N Factors with Decision Rule 'means' and Eigen Type 'EFA': 7



```
# PCA with Scree Plot + Eigenvalues
comp1 <- prcomp(df, scale. = TRUE)

screeplot(comp1, type = 'lines', col = 'red', lwd = 2, pch = 19, cex = 1.2,
           main = 'Scree Plot of HMS Indicators')
```

### Scree Plot of HMS Indicators



```
eigenvalues <- comp1$sdev^2
round(eigenvalues[1:10], 2)
```

```
[1] 11.43  3.36  2.20  1.18  0.92  0.88  0.82  0.62  0.59  0.58
```

Observe that the PCA-based decision rule suggests 4 factors (this is where the real eigenvalues drop below the simulated 95th percentile threshold). Meanwhile, SMC and EFA suggest 11 and 7, respectively. We turn to the scree plot, which has a distinct elbow at 4 factors. Considering after 4, our eigenvalues drop below 1.0, as well as the fact that our data come from four distinct measurement groups (PHQ, GAD, etc.), it makes the most sense to proceed with 4 latent factors.

## Part 5 | Factor Analyses

We now perform a series of factor analyses using orthogonal models with 4 latent factors. We elect to compare all four presented extraction methods: Principal Components, Principal Axis Factoring, Iterative PCA, and Maximum Likelihood. We will evaluate each method based on their root mean squared residual (RSMR), focusing on the percentage of residuals that exceed 0.05.

```
# Principal Components Analysis (PCA)
fact_pca <- principal(df, nfactors = 4, rotate = "varimax")
repro_pca <- fact_pca$loadings %*% t(fact_pca$loadings)
resid_pca <- cor(df) - repro_pca
len_pca <- length(resid_pca[upper.tri(resid_pca)])
rmsr_pca <- sqrt(sum(resid_pca[upper.tri(resid_pca)]^2) / len_pca)
prop_pca <- sum(abs(resid_pca[upper.tri(resid_pca)]) > 0.05) / len_pca * 100

# Principal Axis Factoring (PAF)
fact_paf <- fa(df, nfactors = 4, rotate = "varimax", fm = "pa")
repro_paf <- fact_paf$loadings %*% t(fact_paf$loadings)
resid_paf <- cor(df) - repro_paf
len_paf <- length(resid_paf[upper.tri(resid_paf)])
rmsr_paf <- sqrt(sum(resid_paf[upper.tri(resid_paf)]^2) / len_paf)
prop_paf <- sum(abs(resid_paf[upper.tri(resid_paf)]) > 0.05) / len_paf * 100

# Iterative PCA
fact_ipca <- fa(df, nfactors = 4, rotate = "varimax", SMC = FALSE, fm = "pa")
repro_ipca <- fact_ipca$loadings %*% t(fact_ipca$loadings)
resid_ipca <- cor(df) - repro_ipca
```

```

len_ipca <- length(resid_ipca[upper.tri(resid_ipca)])
rmsr_ipca <- sqrt(sum(resid_ipca[upper.tri(resid_ipca)]^2) / len_ipca)
prop_ipca <- sum(abs(resid_ipca[upper.tri(resid_ipca)]) > 0.05) / len_ipca * 100

# Maximum Likelihood
fact_ml <- factanal(df, factors = 4, rotation = "varimax")
repro_ml <- fact_ml$loadings %*% t(fact_ml$loadings)
resid_ml <- cor(df) - repro_ml
len_ml <- length(resid_ml[upper.tri(resid_ml)])
rmsr_ml <- sqrt(sum(resid_ml[upper.tri(resid_ml)]^2) / len_ml)
prop_ml <- sum(abs(resid_ml[upper.tri(resid_ml)]) > 0.05) / len_ml * 100

```

Table 1: Comparison of Factor Extraction Methods (4-Factor Model)

| Extraction Method                     | RMSR          | % Residuals > 0.05 |
|---------------------------------------|---------------|--------------------|
| Maximum Likelihood (ML)               | 0.0269        | 7.59%              |
| <b>Principal Axis Factoring (PAF)</b> | <b>0.0260</b> | <b>6.67%</b>       |
| <b>Iterative PCA</b>                  | <b>0.0260</b> | <b>6.67%</b>       |
| Principal Components (PCA)            | 0.0392        | 14.25%             |

We see that PCA performs the worst, yielding the highest RMSR of 0.0392 and a substantial 14.25% of residuals above the 0.05 threshold. This matches our expectations, given PCA analyzes total variance instead of isolating shared latent variance. Meanwhile, maximum likelihood yielded improved performance, with an RMSR of 0.0269 and only 7.59% above the threshold.

Both principal axis factoring and iterative PCA converged to the same limits, and thus, they both are our strongest performers, with an RMSR of 0.0260 and 6.67% > 0.05. We elect to proceed with principal axis factoring given its more commonplace usage with psychological constructs like ours, but we would feel equally comfortable continuing with iterative PCA.

## Part 6 | Rotations

Now, we explore two rotational methods: varimax, which maximizes the variance of squared loadings (encouraging each latent factor to be comprised of distinctly correlated questions), and quartimax, which minimizes the number of factors needed to explain each variable (encouraging incorporation of as many questions as possible on to the same latent factor).

```
# Varimax
paf_varimax <- fa(df, nfactors = 4, rotate = 'varimax', fm = 'pa')
# Quartimax
paf_quartimax <- fa(df, nfactors = 4, rotate = 'quartimax', fm = 'pa')
```

Table 2: Comparison of Varimax and Quartimax Factor Loadings

|         | Varimax Rotation |        |       |       | Quartimax Rotation |        |       |       |
|---------|------------------|--------|-------|-------|--------------------|--------|-------|-------|
|         | PA1              | PA2    | PA3   | PA4   | PA1                | PA2    | PA3   | PA4   |
| phq9_1  |                  | 0.659  |       |       | 0.598              | -0.305 |       | 0.374 |
| phq9_2  |                  | 0.635  | 0.322 |       | 0.664              | -0.301 |       | 0.301 |
| phq9_3  |                  | 0.607  |       |       | 0.589              |        |       | 0.324 |
| phq9_4  |                  | 0.653  |       |       | 0.643              |        |       | 0.346 |
| phq9_5  |                  | 0.599  |       |       | 0.604              |        |       | 0.304 |
| phq9_6  | -0.3             | 0.535  | 0.367 |       | 0.643              | -0.307 |       |       |
| phq9_7  |                  | 0.571  | 0.339 |       | 0.627              |        |       |       |
| phq9_8  |                  | 0.451  | 0.311 |       | 0.526              |        |       |       |
| phq9_9  |                  | 0.411  |       |       | 0.417              | -0.306 |       |       |
| gad7_1  |                  | 0.301  | 0.732 |       | 0.797              |        |       |       |
| gad7_2  |                  |        | 0.815 |       | 0.861              |        |       |       |
| gad7_3  |                  | 0.311  | 0.787 |       | 0.848              |        |       |       |
| gad7_4  |                  | 0.383  | 0.676 |       | 0.791              |        |       |       |
| gad7_5  |                  | 0.365  | 0.576 |       | 0.69               |        |       |       |
| gad7_6  |                  | 0.415  | 0.477 |       | 0.65               |        |       |       |
| gad7_7  |                  | 0.333  | 0.587 |       | 0.696              |        |       |       |
| diener1 | 0.771            |        |       |       |                    | 0.78   |       |       |
| diener2 | 0.655            |        |       |       |                    | 0.655  |       |       |
| diener3 | 0.696            | -0.344 |       |       | -0.345             | 0.71   |       |       |
| diener4 | 0.733            |        |       |       |                    | 0.734  |       |       |
| diener5 | 0.746            |        |       |       |                    | 0.751  |       |       |
| diener6 | 0.795            |        |       |       |                    | 0.796  |       |       |
| diener7 | 0.722            |        |       |       |                    | 0.729  |       |       |
| diener8 | 0.678            |        |       |       |                    | 0.676  |       |       |
| BRS_1   |                  |        |       | 0.652 |                    |        | 0.61  |       |
| BRS_2   |                  |        |       | 0.593 | -0.374             |        | 0.53  |       |
| BRS_3   |                  |        |       | 0.675 |                    |        | 0.633 |       |
| BRS_4   |                  |        |       | 0.711 | -0.352             |        | 0.655 |       |
| BRS_5   |                  |        |       | 0.515 |                    |        | 0.477 |       |
| BRS_6   |                  |        |       | 0.676 | -0.365             |        | 0.613 |       |

Observe under quartimax that for PA1, almost every single phq9 and gad7 question was loaded, combining depression and anxiety into the same latent factor. As a result, we see that PA4 has become almost useless, explaining only 3% of the variance. Meanwhile, for the varimax loadings, we see largely separable columns:

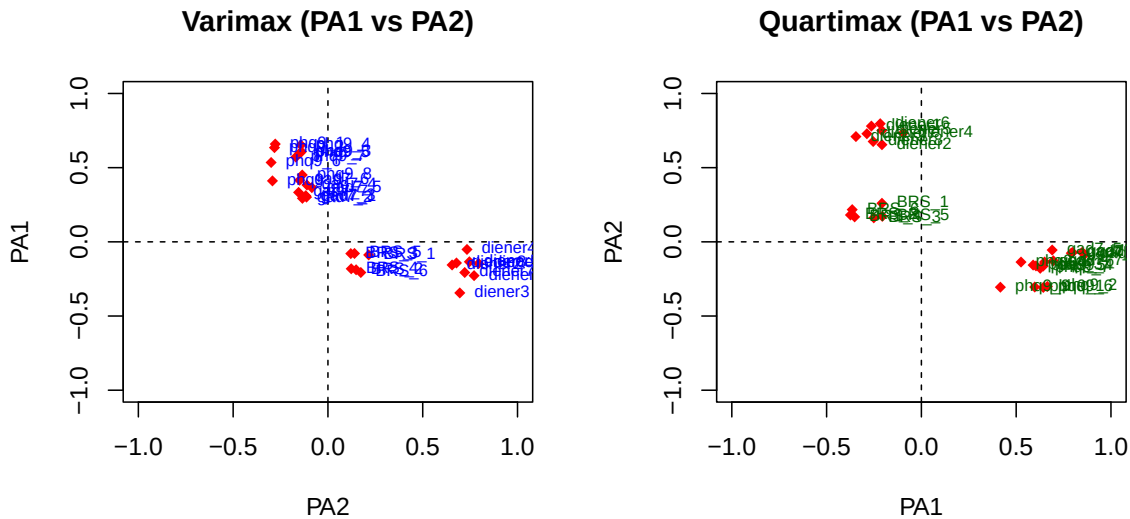
- PA1 contains the flourishing (**diener**) questions
- PA2 contains the **phq9** (depression) questions
- PA3 contains the **gad7** (anxiety) questions, and
- PA4 contains the BRS (resilience) questions.

As expected, there is overlap between PA2 and PA3 given the comorbidity of depression and anxiety.

```
par(mfrow = c(1,2))

# Varimax
plot(paf_varimax$loadings[, c(1, 2)], pch = 18, col = 'red',
     main = "Varimax (PA1 vs PA2)", xlim = c(-1, 1), ylim = c(-1, 1))
abline(h = 0, lty = 2); abline(v = 0, lty = 2)
text(paf_varimax$loadings[, c(1, 2)], labels = rownames(paf_varimax$loadings),
     cex = 0.7, pos = 4, col = "blue")

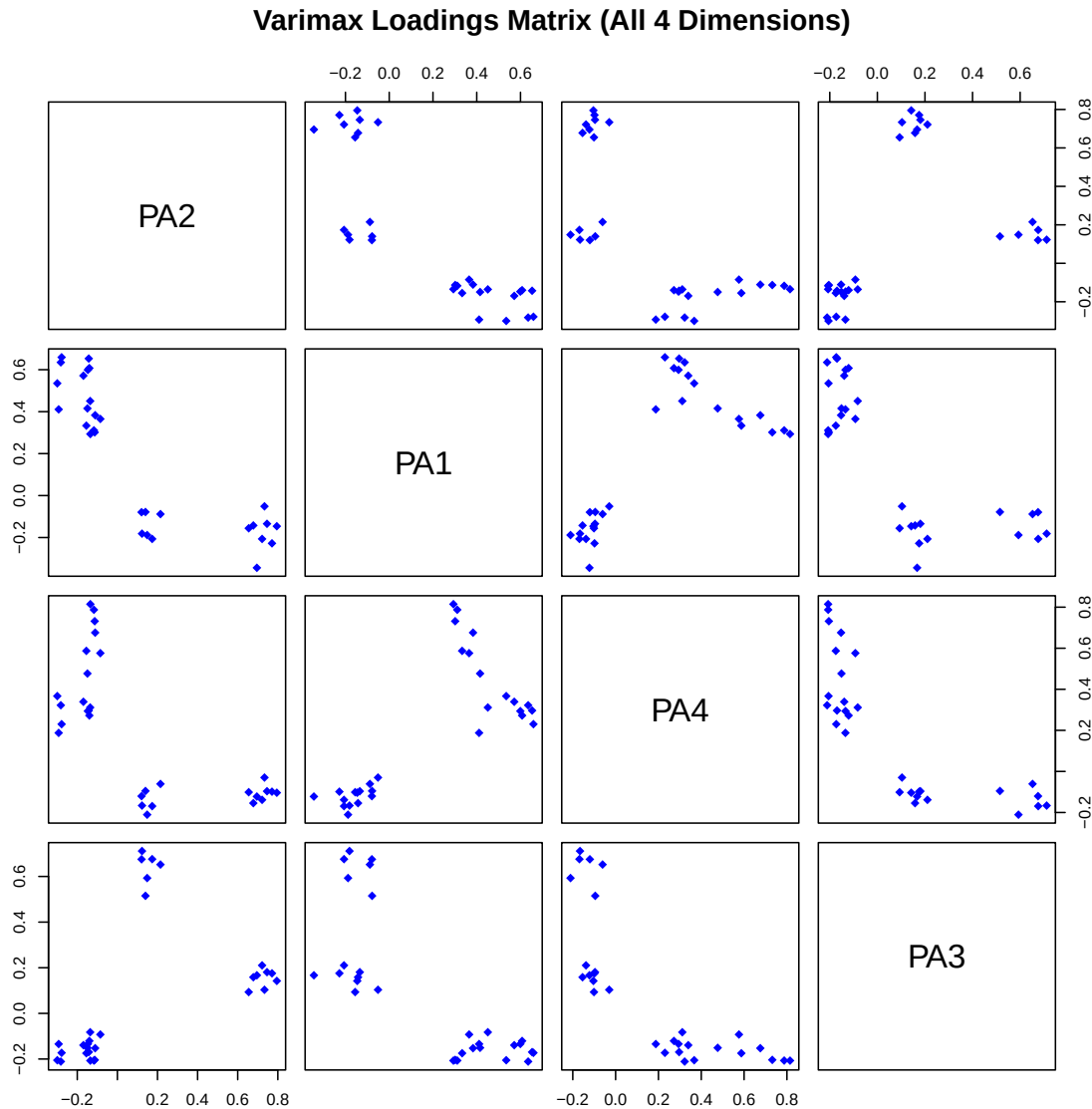
# Quartimax
plot(paf_quartimax$loadings[, c(1, 2)], pch = 18, col = 'red',
     main = "Quartimax (PA1 vs PA2)", xlim = c(-1, 1), ylim = c(-1, 1))
abline(h = 0, lty = 2); abline(v = 0, lty = 2)
text(paf_quartimax$loadings[, c(1, 2)], labels = rownames(paf_quartimax$loadings),
     cex = 0.7, pos = 4, col = "darkgreen")
```



The loading plots confirm this distinction. In the varimax plot of PA1 versus PA2, the variable clusters tightly hug the axes (e.g., flourishing questions load along the vertical axis). In

contrast, the quartimax plot shows variables floating somewhat ambiguously within the quadrants, visually demonstrating cross-loadings across multiple dimensions. Thus, we proceed with the varimax method, presenting all four dimensions via a pairs plot:

```
pairs(paf_varimax$loadings,
      main = "Varimax Loadings Matrix (All 4 Dimensions)",
      pch = 18, col = "blue", cex = 1.2)
```



We see groupings of questions sitting along the vertical and horizontal axes, indicating clear

separation and simple structure. Even with both depression (PA2) and anxiety (PA3), which we know are correlated variables, we see that these have been separated into distinct dimensions, give or take some slight dispersion that reflects the known comorbidity between the two conditions.

## **Part 7 | Summary**

Our exploratory factor analysis successfully identified the latent constructs driving mental health and well-being indicators among the students represented in the 2024–25 Healthy Minds Survey. Initial diagnostics indicated excellent sampling adequacy, with an overall KMO of 0.959. Subsequently, parallel analysis indicated the presence of four distinct latent factors. A comparison of four orthogonal models via RMSR and two rotational methods brought us to decide to utilize principal axis factoring with a varimax rotation.

This approach cleanly resolved our 30 questions into four distinct dimensions that directly map to the questions coded categories: depression, flourishing, anxiety, and resilience. While slight cross-loadings between anxiety and depression reflect their known clinical comorbidity, the model demonstrates that students' mental health profiles are well structured along these four distinct axes of distress and well-being.